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Answer ALL TWENTY SIX questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 Find the lowest common multiple (LCM) of 45 and 70



$$45 = 3^2 \times 5$$

$$70 = 2 \times 5 \times 7$$

$$\text{LCM} : 2 \times 3^2 \times 5 \times 7$$

(Total for Question 1 is 2 marks)



2 The length of a ship is 142.8 m, correct to 1 decimal place.

(i) Write down the lower bound of the length of the ship.

142.75 m
(1)

(ii) Write down the upper bound of the length of the ship.

142.85 m
(1)

(Total for Question 2 is 2 marks)

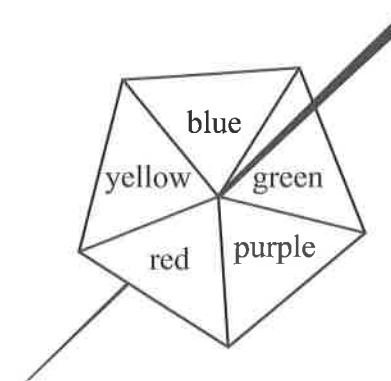
3 Show that $2\frac{1}{4} \times 1\frac{5}{7} = 3\frac{6}{7}$

$$\begin{aligned} & \frac{9}{4} \times \frac{12}{7} \\ &= \frac{108}{28} \\ &= \frac{54}{14} = \frac{27}{7} \\ &= 3\frac{6}{7} \end{aligned}$$

(Total for Question 3 is 3 marks)

4 Here is a biased 5-sided spinner.

When the spinner is spun, it can land on blue or on green or on purple or on red or on yellow.



The table gives information about the probability of the spinner landing on each colour.

Colour	blue	green	purple	red	yellow
Probability	0.12	0.20	0.38	$4x$	x

Sophie spins the spinner once.

(a) Work out the probability that the spinner lands on blue or on green or on purple.

$$\begin{aligned} 0.12 + 0.2 + 0.38 + 4x + x &= 1 \\ 0.7 + 5x &= 1 \\ 5x &= 0.3 \\ x &= \frac{0.3}{5} = 0.06 \\ P(\text{blue or green or purple}) &= 0.12 + 0.20 + 0.38 = 0.70 \end{aligned}$$

Max spins the spinner 350 times.

(b) Work out an estimate for the number of times the spinner lands on red.

$$\begin{aligned} \text{Estimate of Red} &= 350 \times 4(0.06) \\ &= 350 \times 0.24 \\ &= 84 \end{aligned}$$

(4)

(Total for Question 4 is 5 marks)

5 $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

(a) List the members of the set

(i) $A \cup B$

2, 3, 4, 6, 8, 9, 10, 12

(ii) B'

1, 2, 4, 5, 7, 8, 10, 12
(2)

$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

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$\mathcal{E} \quad \cap \quad \cup \quad \emptyset \quad \in \quad \notin$

(b) Write a symbol from the box on each dotted line to make each of the following a true statement.

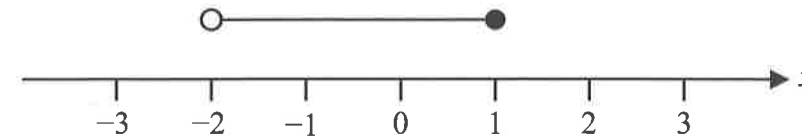
(i) $A \cap C = \emptyset$

(ii) $13 \notin \mathcal{E}$

(2)

(Total for Question 5 is 4 marks)

6



(a) Write down the inequality shown on the number line.

$-2 < x \leq 1$

(2)

(b) Solve the inequality $7a - 5 \leq 3a + 28$
Show clear algebraic working.

$7a - 3a \leq 28 + 5$

$4a \leq 33$

$a \leq 33/4$

(2)

(Total for Question 6 is 4 marks)

7 Change a speed of $50x$ metres per second to a speed in kilometres per hour.

$50x \times 3600$

1000

$50x \times 3.6$

$= 180x$

km/h

(Total for Question 7 is 3 marks)



5 $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

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$C = \{1, 3, 5, 7, 9, 11\}$

(a) List the members of the set

(i) $A \cup B$

2, 3, 4, 6, 8, 9, 10, 12

(ii) B'

1, 2, 4, 5, 7, 8, 10, 12
(2)

$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

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$\mathcal{E} \quad \cap \quad \cup \quad \emptyset \quad \in \quad \notin$

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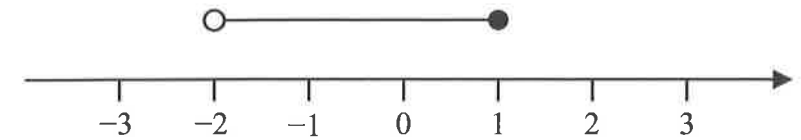
(i) $A \cap C = \emptyset$

(ii) $13 \notin \mathcal{E}$

(2)

(Total for Question 5 is 4 marks)

6



(a) Write down the inequality shown on the number line.

$-2 < x \leq 1$

(2)

(b) Solve the inequality $7a - 5 \leq 3a + 28$
Show clear algebraic working.

$7a - 3a \leq 28 + 5$

$4a \leq 33$

$a \leq 33/4$

(2)

(Total for Question 6 is 4 marks)

7 Change a speed of $50x$ metres per second to a speed in kilometres per hour.

$50x \times \frac{3600}{1000}$

$50x \times 3.6$
 $= 180x$

km/h

(Total for Question 7 is 3 marks)



8 (a) Simplify $a^6 \times a^{10}$

a^{16}

(1)

(b) Simplify $c^{30} \div c^{12}$

c^{18}

(1)

(c) (i) Factorise $y^2 - 10y + 21$

$(y-7)(y-3)$

(2)

(ii) Hence, solve $y^2 - 10y + 21 = 0$

$(y-7)(y-3) = 0$

$y-7=0$

$y=7$

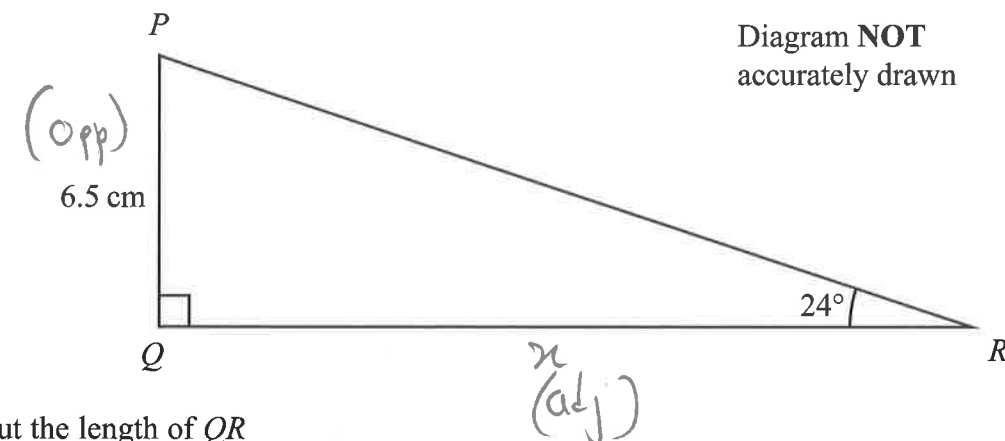
$y-3=0$

$y=3$

(1)

(Total for Question 8 is 5 marks)

9 The diagram shows triangle PQR



Work out the length of QR

Give your answer correct to 3 significant figures.

$$\tan 24 = \frac{6.5}{x}$$

$$x = \frac{6.5}{\tan 24}$$

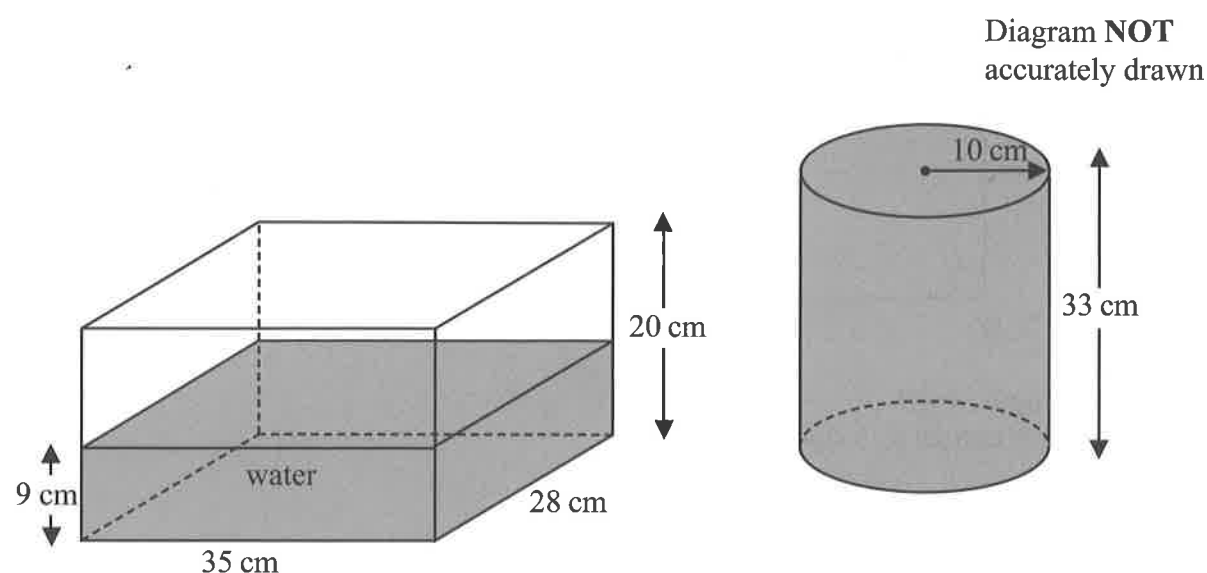
$$= \underline{14.6 \text{ cm}}$$

cm

(Total for Question 9 is 3 marks)



- 10 The diagram shows two water containers.
One is a cuboid and one is a cylinder.



The cuboid measures 35 cm by 28 cm by 20 cm
The surface of the water in the cuboid is 9 cm above the base of the cuboid.

The cylinder has a radius of 10 cm and a height of 33 cm
The cylinder is completely full of water.

Izzy is going to pour all the water from the cylinder into the cuboid.

Show that the cuboid will not be completely full of water.

$$\begin{aligned} \text{Volume of water in cuboid} &= 35 \times 28 \times 9 \\ &= 8820 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Volume of cuboid} &= 35 \times 28 \times 20 \\ &= 19,600 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Unfilled amount} &= 19,600 - 8820 \\ &= 10,780 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Volume of water in cylinder } V &= \pi r^2 h \\ V &= \pi \times 10^2 \times 33 \\ &= 10,367.26 \text{ cm}^3 \end{aligned}$$

Since $10,780 > 10,367.26$
the cuboid will not be completely full.

(Total for Question 10 is 3 marks)

- 11 Zhou invests some money for 2 years.
He invests \$2500 with Bank A and \$3000 with Bank B.

Bank A	
Invests \$2500	
amount of money invested	total amount of interest after 2 years = 20 : 3

Bank B	
Invests \$3000	
4% per year compound interest for 2 years	

Zhou receives more **interest** from Bank A than from Bank B.

How much more?

Bank B

$$\begin{aligned} V &= P \left(1 + \frac{r}{100}\right)^n \\ V &= 3000 \left(1 + \frac{4}{100}\right)^2 \\ V &= \$3244.80 \end{aligned}$$

$$\begin{aligned} \text{Interest} &= 3244.80 - 3000 \\ &= \$244.80 \end{aligned}$$

How much more

$$\begin{aligned} &300 - 244.80 \\ &= \$55.20 \end{aligned}$$

Bank A

$$\begin{aligned} \text{Invests } \$2300 \\ 20:3 &= 23 \text{ parts} \\ 1 \text{ part} &= \frac{2300}{23} = \$100 \end{aligned}$$

$$\begin{aligned} &\times 10 \quad \begin{matrix} 20:3 \\ \downarrow \end{matrix} \quad \times 100 \\ &2000 : 300 \\ \text{Interest} &= \$300 \end{aligned}$$

$$\$55.20$$

(Total for Question 11 is 5 marks)

12 The table gives information about the ages of 80 people in a cinema.

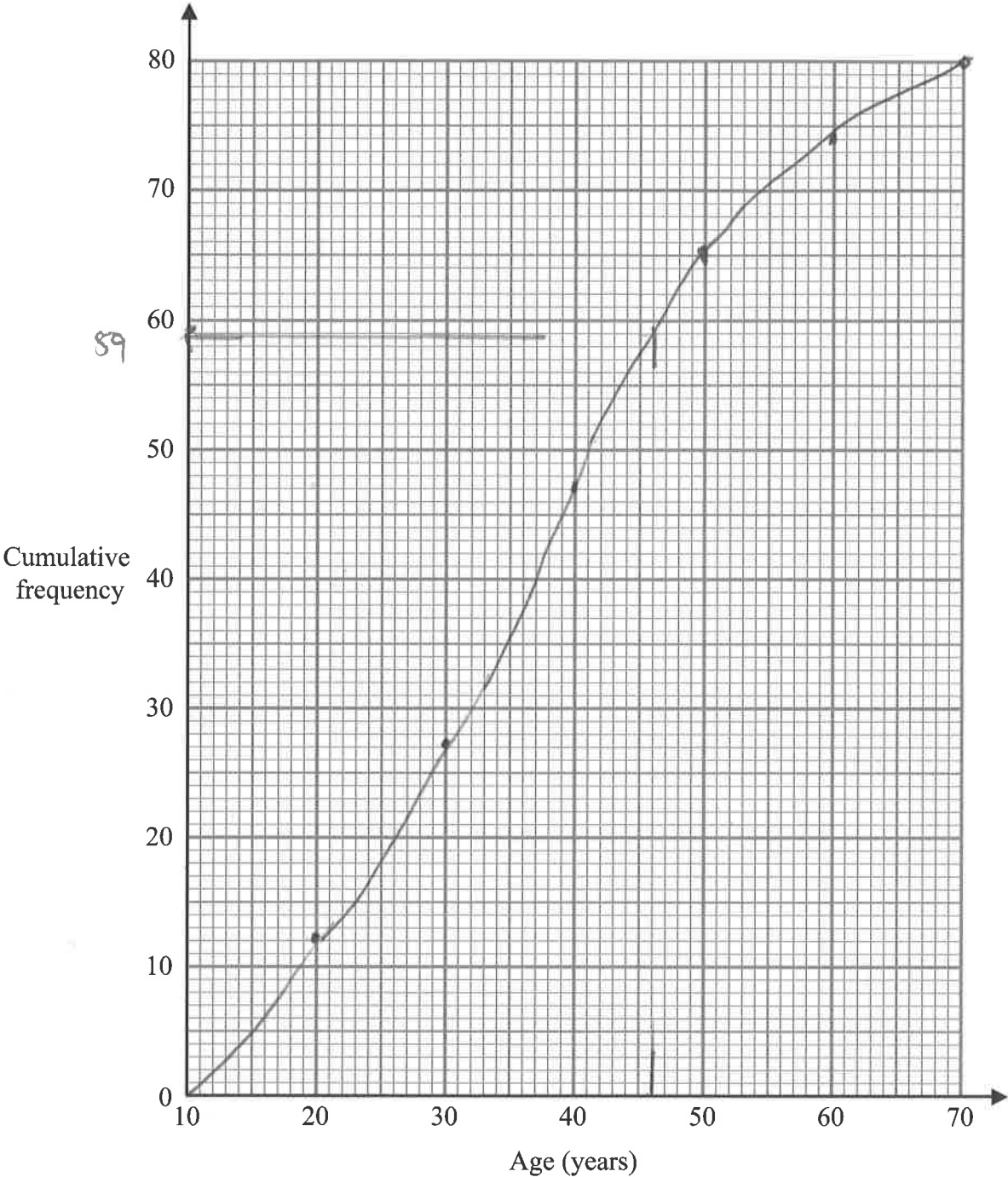
Age (n years)	Frequency
$10 < n \leq 20$	12
$20 < n \leq 30$	15
$30 < n \leq 40$	20
$40 < n \leq 50$	18
$50 < n \leq 60$	9
$60 < n \leq 70$	6

(a) Complete the cumulative frequency table.

Age (n years)	Cumulative frequency
$10 < n \leq 20$	12
$10 < n \leq 30$	27
$10 < n \leq 40$	47
$10 < n \leq 50$	65
$10 < n \leq 60$	74
$10 < n \leq 70$	80

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the percentage of the 80 people who are more than 46 years of age.
Give your answer correct to the nearest whole number.

$$\frac{80 - 59}{80} \times 100$$
$$\frac{21}{80} \times 100 =$$

26 %
(3)

(Total for Question 12 is 6 marks)



13 Here are the numbers of eggs laid by some hens on each of 11 days in June.

9 10 12 15 17 18 19 19 20 25 26

LQ

Median

UQ

Work out the interquartile range of the numbers of eggs.

$$\begin{aligned} & \text{UQ} - \text{LQ} \\ &= 20 - 12 \\ &= 8 \end{aligned}$$

(Total for Question 13 is 2 marks)

14 Work out $6.7 \times 10^{135} + 3 \times 10^{134}$

Give your answer in standard form.

$$\begin{array}{r} 67, \text{ } \\ + 3, \text{ } \\ \hline 70, \text{ } \end{array}$$

$$70 \times 10^{134}$$

$$7.0 \times 10^{135}$$

(Total for Question 14 is 2 marks)

15 (a) Solve $\frac{5a+8}{3} - \frac{2a+5}{4} = 23$

Show clear algebraic working.

$$\begin{aligned} & \frac{5a+8}{3} - \frac{2a+5}{4} = 23 \\ & \frac{4(5a+8) - 3(2a+5)}{12} = 23 \end{aligned}$$

$$20a + 32 - 6a - 15 = 23 \times 12$$

$$14a + 17 = 276$$

$$14a = 276 - 17$$

$$14a = 259$$

$$a = \frac{259}{14}$$

$a =$ (4)

(b) Express $\left(\frac{\sqrt{y}}{3}\right)^{-1}$ in the form cy^n where c and n are numbers to be found.

$$\left(\frac{y^{\frac{1}{2}}}{3}\right)^{-1} = \left(\frac{3}{y^{\frac{1}{2}}}\right)^1 = \frac{3}{y^{\frac{1}{2}}} = 3y^{-\frac{1}{2}}$$

(2)

(Total for Question 15 is 6 marks)



- 16 Use algebra to show that the recurring decimal $0.6\dot{1}\dot{2} = \frac{101}{165}$

$$\begin{aligned}
 x &= 0.6\dot{1}\dot{2} \\
 (100x &= 61.2\dot{1}\dot{2}) \\
 99x &= 60.6 \\
 x &= \frac{60.6}{99} \\
 x &= \frac{606}{990} \\
 &= \frac{101}{165}
 \end{aligned}$$

(Total for Question 16 is 2 marks)

- 17 Prove that, for any three numbers which are consecutive multiples of 4, the difference between the square of the largest number and the square of the smallest number is always a multiple of 64

Show clear algebraic working.

$$\begin{aligned}
 &4n, 4n+4, 4n+8 \\
 &(4n+8)^2 - (4n)^2 \\
 &16n^2 + 64n + 64 - 16n^2 \\
 &64n + 64 \\
 &64(n+1)
 \end{aligned}$$

Since we can factor out 64,
this shows it is a multiple of 64.

(Total for Question 17 is 3 marks)



18 F is inversely proportional to the cube of r

$F = 6$ when $r = 2$

(a) Find a formula for F in terms of r

$$F \propto \frac{1}{r^3}$$

$$F = \frac{K}{r^3}$$

$$6 = \frac{K}{2^3}$$

$$6 \times 2^3 = K$$

$$48 = K$$

$$F = \frac{48}{r^3}$$

(b) Find the value of r when $F = 3072$

$$3072 = \frac{48}{r^3}$$

$$r^3 = \frac{48}{3072}$$

$$r = \sqrt[3]{\frac{48}{3072}}$$

$$= \frac{1}{4}$$

$r =$ (2)

(Total for Question 18 is 5 marks)

19 Kannika has 9 counters.

There is a number on each counter.



Kannika puts the 9 counters in a bag.

She takes at random a counter from the bag and does not replace the counter.

She then takes at random a second counter from the bag.

Work out the probability that the sum of the numbers on the two counters is less than 5

Sum is less than 5

$(1, 1), (1, 2), (1, 3_A)$

$(1_B, 2), (1_A, 3_B)$

$(1_A, 3_C)$

$(1_B, 3_A)$

$(1_B, 3_B)$

$(1_B, 3_C)$

1 way 2 ways 6 ways

$$\text{Probability} = \frac{2}{9} \times \frac{1}{8} + \left(\frac{2}{9} \times \frac{1}{8} \right) \times 2 + \frac{2}{9} \times \frac{3}{8} + \frac{3}{9} \times \frac{2}{8}$$

$$= \frac{2}{72} + \frac{4}{72} + \frac{6}{72} + \frac{6}{72} = \frac{18}{72} = \frac{1}{4}$$

(Total for Question 19 is 3 marks)



20 The diagram shows cuboid $ABCDEFGH$

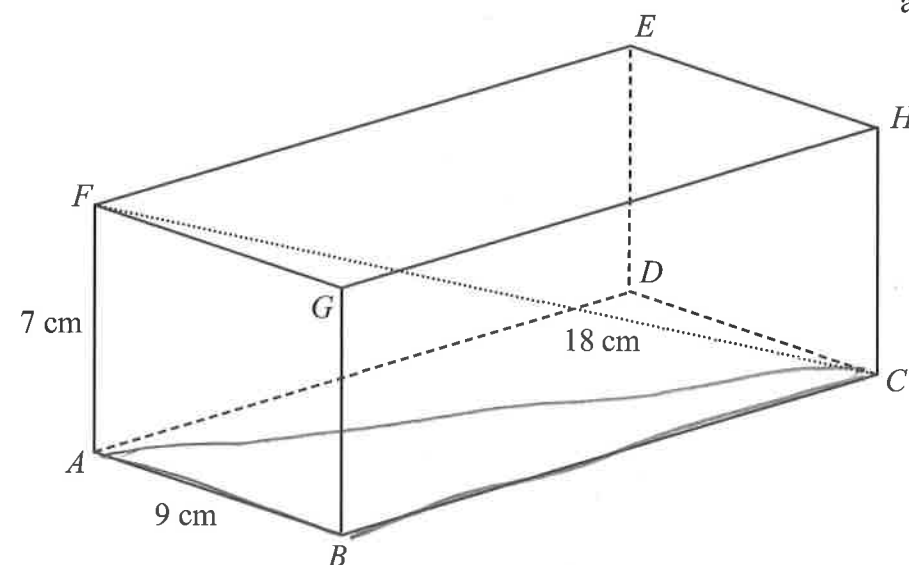


Diagram **NOT** accurately drawn

$$AB = 9 \text{ cm} \quad AF = 7 \text{ cm} \quad FC = 18 \text{ cm}$$

Calculate the length of BC
Give your answer correct to 3 significant figures.

$$AC = \sqrt{18^2 - 7^2} \\ = 5\sqrt{11}$$

$$BC = \sqrt{(5\sqrt{11})^2 - 9^2} \\ = \sqrt{194} \\ = 13.9 \text{ cm}$$

(Total for Question 20 is 3 marks)

21 The curve with equation $y = f(x)$ has one turning point.
The coordinates of this turning point are $(-6, 9)$

- (a) Write down the coordinates of the turning point on the curve with equation $y = f(3x)$

$$\left(\frac{-2}{1}, \frac{9}{1} \right)$$

The curve C with equation $y = g(x)$ is transformed to give the curve S with equation $y = g(x + a) + b$

The point $(4, -5)$ on C is mapped to the point $(1, -16)$ on S

- (b) Write down the value of a and the value of b

$$a = 3 \\ b = -11 \\ (2)$$

(Total for Question 21 is 3 marks)



22 Solve the simultaneous equations

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

Show clear algebraic working.

$$y = 3 - 2x$$

$$x^2 + y^2 = 41$$

$$x^2 + (3 - 2x)^2 = 41$$

$$x^2 + 9 - 12x + 4x^2 = 41$$

$$5x^2 - 12x + 9 - 41 = 0$$

$$5x^2 - 12x - 32 = 0 \quad \times (-160)$$

$$5x^2 - 20x + 8x - 32 = 0 \quad + (-12)$$

$$5x(x - 4) + 8(x - 4) = 0$$

$$(5x + 8)(x - 4) = 0$$

$$5x + 8 = 0 \quad x - 4 = 0$$

$$x = -\frac{8}{5} \quad x = 4$$

$$y = 3 - 2(-\frac{8}{5}) \quad y = 3 - 2(4)$$

$$= 3 + \frac{16}{5} \quad y = 3 - 8$$

$$= \frac{31}{5} \quad y = -5$$

==

(Total for Question 22 is 5 marks)

23

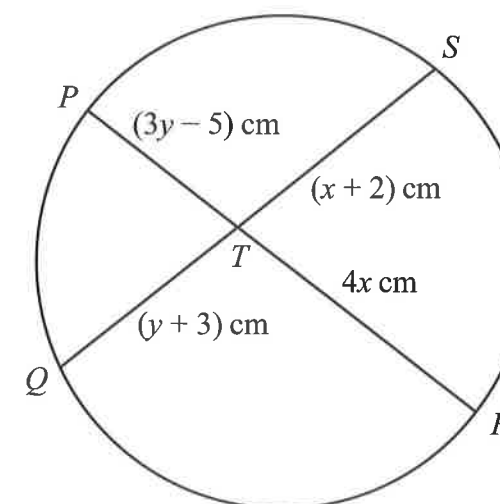


Diagram NOT accurately drawn

PTR and QTS are chords of a circle.

$$PT = (3y - 5) \text{ cm} \quad QT = (y + 3) \text{ cm} \quad RT = 4x \text{ cm} \quad ST = (x + 2) \text{ cm}$$

Find an expression for y in terms of x

$$4x(3y - 5) = (y + 3)(x + 2)$$

$$12xy - 20x = xy + 2y + 3x + 6$$

$$12xy - xy - 2y = 3x + 20x + 6$$

$$11xy - 2y = 23x + 6$$

$$y(11x - 2) = 23x + 6$$

$$y = \frac{23x + 6}{11x - 2}$$

y =

Total for Question 23 is 5 marks)



24 The first 3 terms of an arithmetic series are

$$(2x + 5) \quad (3y - 4) \quad (4x - 2)$$

where x and y are constants.

The sum of the first 9 terms of the series is 216

Find the value of x and the value of y
Show clear algebraic working.

Common difference

$$3y - 4 - (2x + 5) = 4x - 2 - (3y - 4)$$

$$3y - 4 - 2x - 5 = 4x - 2 - 3y + 4$$

$$3y - 2x - 9 = 4x - 3y + 2 \quad \text{each represents 'd'}$$

$$6y - 9 - 2 = 4x + 2x$$

$$6y - 11 = 6x \quad \dots (1)$$

$$S_9 = \frac{9}{2} [2(2x + 5) + (9 - 1)(4x - 3y + 2)] = 216$$

$$= \frac{9}{2} [4x + 10 + 32x - 24y + 16] = 216$$

$$\Rightarrow 36x - 24y + 26 = 216 \times \frac{2}{9}$$

$$36x - 24y + 26 = 48$$

Using equation (1) $6x = 6y - 11$

$$\text{So } 36x = 36y - 66 \quad (\times 6)$$

Now substituting

$$36y - 66 - 24y + 26 = 48$$

$$12y - 40 = 48$$

$$12y = 48 + 40$$

$$12y = 88$$

$$y = \frac{88}{12} = \frac{22}{3}$$

$$6y - 11 = 6x$$

$$6\left(\frac{22}{3}\right) - 11 = 6x$$

$$44 - 11 = 6x$$

$$33 = 6x$$

$$\frac{33}{6} = x$$

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 24 is 6 marks)

Turn over for Question 25



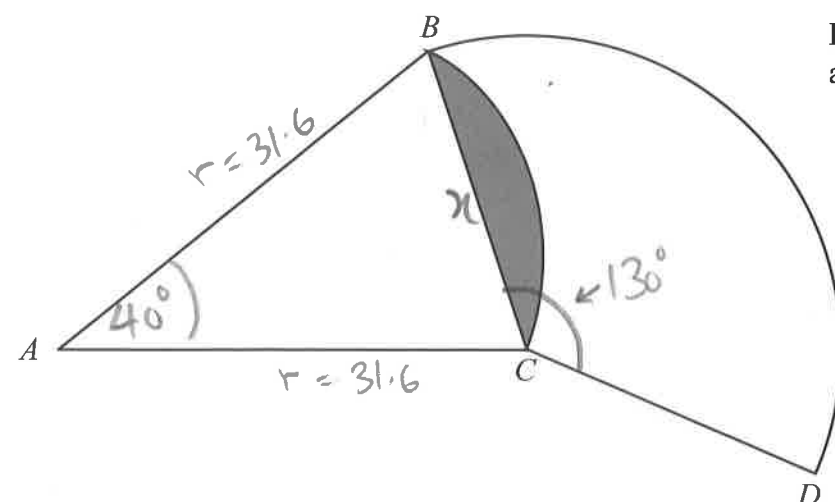


Diagram NOT
accurately drawn

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BAC is a sector of a circle, centre A

BCD is a sector of a circle, centre C

Angle $BAC = 40^\circ$

Angle $BCD = 130^\circ$

Area of shaded segment = 28 cm^2

Find the length of the arc BD

Give your answer correct to 3 significant figures.

$$\text{Area of sector} - \text{Area of triangle} = 28$$

$$\frac{40}{360} \times \pi r^2 - \frac{1}{2} \times r \times r \times \sin 40 = 28$$

$$\frac{1}{9} \pi r^2 - 0.321 r^2 = 28$$

$$0.349 r^2 - 0.321 r^2 = 28$$

$$0.028 r^2 = 28$$

$$r^2 = \frac{28}{0.028}$$

$$r = \sqrt{\frac{28}{0.028}}$$

$$= 31.6 \text{ cm}$$

Using Cosine Rule
to find BC .

$$x = \sqrt{31.6^2 + 31.6^2 - 2(31.6)(31.6)\cos 40}$$

$$= 21.6 \text{ cm}$$

$$\text{Length of arc} = \frac{130}{260} \times 2 \times \pi \times 21.6$$

$$= 67.9 \text{ cm}$$

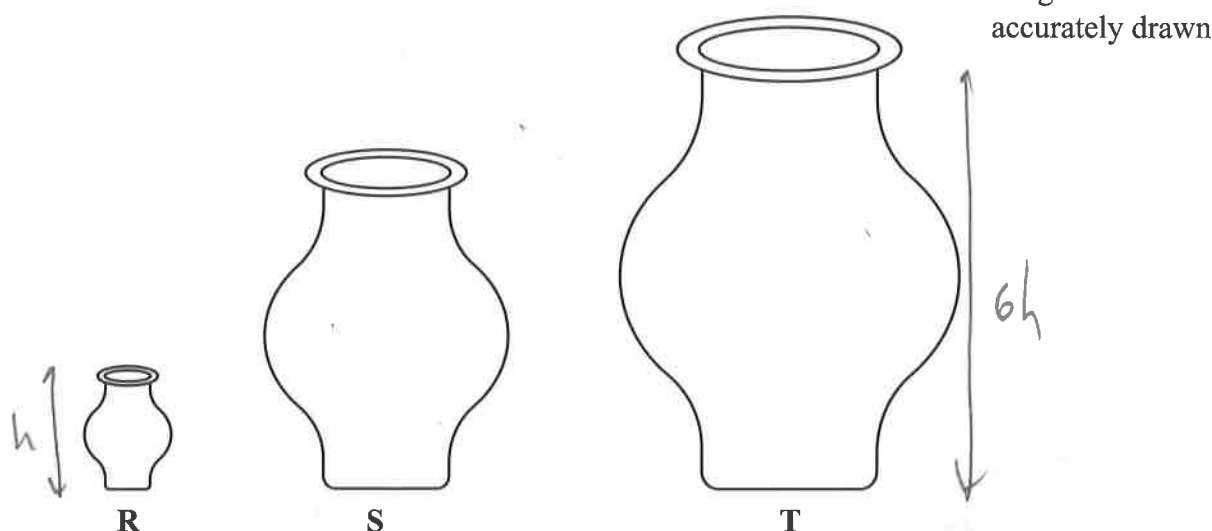
cm

(Total for Question 25 is 6 marks)

Turn over for Question 26



26 R, S and T are three similar vases.



The volume of vase S is 72.8% more than the volume of vase R

$$Vol\ S = 1.728\ Vol\ R$$

The height of vase R is h cm

The height of vase T is $6h$ cm

The surface area of vase S is A cm²

The surface area of vase T is kA cm²

Work out the value of k

$$\text{Volume ratio} = (\text{Scale factor})^3$$

$$x^3 = 1.728$$

$$x = \sqrt[3]{1.728}$$

$$x = 1.2$$

Scale factor from R to S is 1.2

Scale factor from R to T is :

$$\frac{6h}{h} = 6$$

$$\text{Surface area ratio} = (\text{Scale factor})^2$$

$$k = (\text{scale factor from S to T})^2$$

Scale from R to S = 1.2

Scale from R to T = 6

Scale from S to T = $\frac{6}{1.2} = 5$

$$k = 5^2 = 25$$

$$k = 25$$

(Total for Question 26 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS

